Tutorial 2

The objective of this tutorial is to get you familiar with some fundamental commands you can use for modeling, designing and tuning control systems.

Perform all the programming steps in a single file named **<roll_number>_T2.m** and submit that along with a PDF of the answers by **midnight**.

1. For the system ID problem given in class,(i.e. the bode plot of a black box), write a MATLAB script to identify the required salient features needed to generate the transfer function.
2. For the function $G(s) = 16/(s^3+6s^2+5s+8)$, prove through MATLAB and Simulink, that the closed loop system becomes unstable at the gains of the phase margin and beyond for the given bandwidth. Also prove that if you introduce a phase offset equal to the phase margin, the system becomes unstable.